CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Agriculture remains a major contributor to the economy of many developing countries, particularly Nigeria, where a significant proportion of the population depends on farming for livelihood. Among the various agricultural crops cultivated, soybean (Glycine max) has gained considerable importance due to its high protein content, oil yield, and diverse applications in food processing, animal feed production, and industrial manufacturing. The increasing demand for soybean products has necessitated for improvements in post-harvest processing technologies to enhance product quality and reduce losses.

One of the major challenges associated with soybean processing is the presence of impurities such as dust, chaff, stones, broken seeds, and other foreign materials introduced during harvesting, threshing, handling, and storage. These impurities reduce the market value of soybeans, affect processing efficiency, and compromise the quality of finished products. Traditionally, soybean cleaning in rural communities is carried out manually through winnowing and hand sorting, which are labor-intensive, time-consuming, and often inefficient.

Mechanical cleaning systems have been developed to improve the efficiency and effectiveness of grain cleaning operations. However, many commercially available soybean cleaning machines are powered by electricity or fossil fuels, making them expensive and inaccessible to small-scale farmers, particularly in rural areas where

electricity supply is unreliable or unavailable. This challenge has created the need for affordable, sustainable, and energy-efficient alternatives suitable for resource-constrained environments.

Pedal-operated machines provide a practical solution by utilizing human power to generate mechanical energy for operation. Pedal-powered technologies have been successfully applied in various agricultural processing activities due to their simplicity, low operating cost, environmental friendliness, and independence from external energy sources. The development of a pedal-operated soybean cleaning machine offers the potential to reduce labor requirements, improve cleaning efficiency, and increase productivity among smallholder farmers and rural processors.

This study focuses on the development and performance evaluation of a pedal-operated soybean cleaning machine designed to effectively remove impurities from soybean grains while maintaining affordability, simplicity, and operational efficiency. The study also aims to assess the machine's performance characteristics under different operating conditions to determine its suitability for small-scale agricultural applications.

1.2 Statement of the Problem

Soybean production has increased significantly due to its nutritional and economic importance, in post-harvest handling and processing remain the major challenges, particularly in rural communities where access to mechanized equipment is limited. Traditional methods of cleaning soybeans are characterized by low efficiency, high

labor requirements, and inconsistent cleaning performance.

Most existing soybean cleaning machines depend on electrical or fuel-powered systems, which may not be economically viable for small-scale farmers due to high acquisition costs, maintenance expenses, and inadequate access to reliable power supply. Consequently, many farmers continue to rely on manual cleaning methods, resulting in reduced productivity, increased processing time, and lower product quality.

There is, therefore, a need to develop a low-cost, efficient, and environmentally sustainable soybean cleaning machine that can operate independently of electricity while providing satisfactory cleaning performance. This study seeks to address this challenge through the development and evaluation of a pedal-operated soybean cleaning machine.

1.3 Aim of the Study

The aim of this study is to develop a pedal-operated Soyabean cleaning machine suitable for small-scale agricultural processing operations.

1.4 Objectives of the Study

The objectives of this study are to:

1. Determine some physical properties of soya beans relevant to cleaning.
2. Evaluate the performance of the machine using selected performance parameters.

3. Assess the operational suitability and effectiveness of the machine for small-scale soyabean processing.

4. To design the components, fabrication and assembly of the machine

1.5 Research Questions

The study seeks to answer the following questions:

1. Can a soybean cleaning machine be effectively operated using pedal power?

2. What level of cleaning efficiency can be achieved by the developed machine?

3. What is the output capacity of the pedal-operated soybean cleaning machine?

4. How effective is the machine in removing impurities from soybean grains?

5. Is the developed machine suitable for small-scale soybean processing operations?

1.6 Significance of the Study

The development of a pedal-operated soybean cleaning machine is significant for several reasons. First, it provides an affordable and accessible technology for smallholder farmers and rural processors who lack access to electricity or motorized equipment. Second, the machine can reduce the labor intensity associated with traditional soybean cleaning methods while improving productivity and operational efficiency.

The study also contributes to the promotion of appropriate technology development by encouraging the use of locally available materials and human-powered systems.

Furthermore, the findings of this research will provide useful information to agricultural engineers, researchers, policymakers, and equipment manufacturers interested in developing sustainable agricultural processing technologies.

In addition, the successful development of the machine can enhance soybean quality, reduce post-harvest losses, increase farmers' income, and support food security and rural industrialization.

1.7 Scope of the Study

This study is limited to the design, fabrication, and performance evaluation of a pedal -operated soybean cleaning machine. The study focuses on the cleaning of harvested soybean grains through the removal of impurities such as dust, chaff, broken seeds, and other foreign materials. Performance evaluation will be based on parameters such as performance indicator.

The study does not cover large-scale industrial soybean cleaning systems, electrical power systems, or advanced automated grain processing technologies.

1.8 Justification of the Study

The need for affordable and sustainable agricultural processing equipment has become increasingly important due to rising energy costs and limited access to electricity in rural areas. A pedal-operated soybean cleaning machine offers a practical alternative that can operate without external power sources while maintaining acceptable performance levels.

The use of human-powered technology promotes environmental sustainability,

reduces operational costs, and enhances technology accessibility for low-income farmers. Therefore, the development of a pedal-operated soybean cleaning machine is justified as a means of improving post-harvest processing efficiency, reducing labor requirements, and supporting agricultural development in resource-limited communities.

1.9 Definition of Terms

Soybean: A leguminous crop cultivated for its high protein and oil content and used in food, feed, and industrial applications.

Cleaning Efficiency: The percentage of impurities removed from soybean grains during the cleaning process.

Output Capacity: The quantity of soybean cleaned per unit time, usually expressed in kilograms per hour (kg/h).

Pedal Operation: A mechanical system in which human effort applied through pedaling is converted into rotational mechanical energy.

Impurities: Unwanted materials such as dust, chaff, stones, broken grains, and foreign particles present in harvested soybeans.

Performance Evaluation: The process of assessing the operational effectiveness and efficiency of a machine using predetermined performance indicators.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

Soybean (Glycine max L. Merrill) is one of the world's most important grain legumes because of its high protein, oil content, and wide range of industrial and food applications. It serves as a major source of edible oil, livestock feed, and raw material for numerous food products. In Nigeria, soybean cultivation has increased considerably due to its nutritional value and economic importance. However, the quality of harvested soybean is often reduced by the presence of impurities such as dust, stones, chaff, broken pods, weed seeds, and other foreign materials acquired during harvesting, handling, transportation, and storage. These impurities reduce market value, processing efficiency, storage life, and consumer acceptability.

Cleaning is therefore an essential post-harvest operation that improves grain quality before storage, marketing, processing, or planting. Efficient cleaning minimizes quantitative and qualitative losses while increasing the commercial value of soybean. Traditional cleaning methods used by many smallholder farmers are laborious, time-consuming, and less efficient, thereby creating the need for affordable and efficient cleaning technologies suitable for rural communities.

Pedal-operated cleaning machines have emerged as an intermediate technology capable of bridging the gap between manual cleaning and expensive motorized equipment. These machines utilize human power to drive cleaning mechanisms, making them suitable for off-grid rural communities where electricity is unavailable

or unreliable. Consequently, the development of a pedal-operated soybean cleaning machine offers a practical solution for improving post-harvest operations among small- and medium-scale farmers.

2.2 Concept of Soya bean

Soya bean (Glycine max L. Merrill) belongs to the family Fabaceae and is recognized globally as one of the most economically important oilseed crops. The crop originated in East Asia and has become widely cultivated because of its exceptional nutritional composition, containing approximately 35–40% protein and 18–22% oil. Soybean also contains carbohydrates, minerals, vitamins, and essential amino acids that make it valuable for both human consumption and livestock production.

According to the Food and Agriculture Organization of the United Nations (FAO 1996 and FAO 2017), soybean plays a significant role in improving food security and household income due to its versatility in food, feed, and industrial applications. It is processed into soybean flour, soy milk, soybean oil, tofu, animal feed, and numerous industrial products. Soybean cultivation has expanded rapidly across Africa, particularly in Nigeria, where it contributes significantly to agricultural development and employment.

Several researchers have emphasized that maintaining grain quality after harvesting is as important as increasing production. Foreign materials mixed with harvested soybeans reduce storage stability, interfere with processing equipment, increase transportation costs, and lower product quality. Therefore, efficient cleaning remains one of the most important post-harvest operations in soybean processing.

Soybean (Glycine max) is one of the world's most important leguminous crops due to its high nutritional value and diverse industrial applications. Soybeans contain approximately 35–40% protein and 18–20% oil, making them an essential source of food, livestock feed, and industrial raw materials. The crop is widely cultivated in many countries, including Nigeria, where it contributes significantly to food security, income generation, and agricultural industrialization (Toomer et al., 2023).

The increasing demand for soybean products has resulted in expanded cultivation and processing activities. However, efficient post-harvest processing remains a major challenge, especially among smallholder farmers who rely on traditional methods. The quality of soybean products largely depends on effective post-harvest operations such as threshing, drying, cleaning, grading, storage, and transportation (Satimehin et al., 2019).

2.3 Post harvest Processing of Soybeans

Post harvest processing refers to all activities performed after harvesting to preserve product quality and reduce losses. In soybean processing, these activities include drying, threshing, cleaning, sorting, grading, and storage. Proper post-harvest management enhances market value, minimizes deterioration, and improves processing efficiency (Kabir & Fedele, 2018).

Studies have shown that post-harvest losses in soybean production can be substantial due to inadequate processing technologies, poor storage conditions, and inefficient cleaning methods. Traditional cleaning methods such as hand sorting and manual winnowing are characterized by low efficiency, high labor requirements, and

inconsistent cleaning performance (Satimehin et al., 2019).

The cleaning operation is considered one of the most important postharvest processing stages because it removes foreign materials such as dust, stones, chaff, broken grains, and other contaminants. Effective cleaning improves product quality, increases storage life, enhances processing efficiency, and improves market acceptance.

2.4 Concept and Definition of Cleaning

Cleaning is one of the primary post-harvest operations carried out to separate useful agricultural produce from unwanted foreign materials. Different scholars have defined cleaning from various perspectives.

According to Food and Agriculture Organization of the United Nations, cleaning is the process of removing dust, chaff, stones, weed seeds, broken grains, and other impurities from harvested grains in order to improve quality, storage characteristics, and processing efficiency.

Moore (1983) described cleaning as a preparatory operation in soybean processing that removes foreign materials before drying, dehulling, cracking, and oil extraction, thereby improving product quality and protecting processing equipment from damage.

Akatuhurira et al. (2021) defined seed cleaning as the separation of desired seeds from contaminants using differences in physical properties

2.5 Importance of Soya bean Cleaning

Cleaning is one of the most important post-harvest operations in soybean production because it improves grain quality, enhances storage stability, and increases market value. Harvested soybeans usually contain impurities such as stones, soil particles, chaff, broken pods, weed seeds, sticks, leaves, dust, and damaged grains. If these contaminants are not removed, they reduce the quality of the grain, encourage insect infestation and mould growth during storage, and interfere with subsequent processing operations.

The Food and Agriculture Organization (FAO) reported that proper cleaning improves the commercial value of grains and pulses, reduces post-harvest losses, facilitates safe storage, and enhances processing efficiency. Clean grains also attract better market prices because they satisfy grading standards and consumer expectations.

According to Toomer et al. (2023), soybean cleaning is an essential operation before dehulling, cracking, oil extraction, and food processing because foreign materials reduce processing efficiency and may damage processing equipment. The removal of dust, stones, metallic particles, and other contaminants also improves food safety and product quality.

2.6 Types of Soya bean Cleaning Methods

Several methods have been developed for cleaning soybeans depending on the scale of production, available technology, labour availability, and economic considerations. These methods are broadly classified into manual and mechanical cleaning methods.

2.6.1 Manual Cleaning Methods

Manual cleaning is the oldest and most widely practiced method among smallholder farmers in developing countries. It involves hand-picking impurities, winnowing, sieving, and washing where appropriate. These operations depend entirely on human labour and simple tools such as baskets, trays, woven sieves, and shallow containers.

Hand picking is suitable for removing large contaminants such as stones, sticks, damaged pods, and other visible foreign materials. Although the method is inexpensive, it is slow, labour-intensive, and unsuitable for handling large quantities of soybean.

Winnowing is another traditional cleaning technique that separates lighter impurities from soybean seeds using natural or artificial air currents. During winnowing, the grain mixture is poured from a height, allowing heavier soybean seeds to fall vertically while lighter materials such as chaff, dust, and dry leaves are blown away by the wind. The efficiency of winnowing depends largely on wind speed, operator skill, and weather conditions.

Sieving is commonly used after winnowing to remove impurities based on particle size. Different mesh sizes allow soil particles, sand, and broken materials to pass through while retaining soybean grains. Although sieving improves grain cleanliness, it is time-consuming and cannot effectively remove impurities having dimensions similar to soybean seeds.

Despite their simplicity and low cost, manual cleaning methods suffer from low productivity, inconsistent cleaning efficiency, high labour requirements, and operator

fatigue. Consequently, these limitations have encouraged the development of improved mechanical cleaning technologies.

2.6.2 Mechanical Cleaning Methods

Mechanical cleaning utilizes machines to separate soybean grains from impurities according to differences in size, shape, density, weight, and aerodynamic properties. Mechanical cleaners are widely used because they increase throughput, improve cleaning efficiency, reduce labour requirements, and minimize grain losses.

The most common mechanical soybean cleaning machines include vibrating screen cleaners, rotary screen cleaners, aspirator cleaners, gravity separators, air-screen cleaners, and combined cleaning machines.

Screen or Sieve Cleaners: These machines use vibrating or oscillating screens of different mesh sizes to separate impurities according to particle size. Large impurities remain on the upper screen while smaller particles pass through the lower screen, leaving clean soybean grains between the screens.

Aspirator (Air) Cleaners: Aspirator cleaners employ centrifugal fans or blowers to generate airflow that removes light impurities such as dust, husks, leaves, and chaff based on aerodynamic properties. Air cleaning is usually combined with screening to improve overall cleaning performance.

Gravity Separators: Gravity separators classify soybean grains according to differences in density. Heavier and healthy seeds move differently from lighter damaged seeds and foreign materials when subjected to vibration and airflow. These

machines produce high-quality seed suitable for planting and commercial processing.

Rotary Drum Cleaners: Rotary cleaners consist of cylindrical perforated drums rotating at moderate speeds. Materials are separated according to size while moving through the rotating drum. Rotary cleaners are suitable for handling relatively large quantities of grain but may require additional aspiration systems for complete cleaning.

Combined Cleaning Systems: Modern grain cleaners combine screening, aspiration, grading, and sometimes magnetic separation into a single unit. Such integrated systems provide higher cleaning efficiency while minimizing grain losses and increasing processing capacity. Magnetic separators are particularly useful for removing metallic contaminants that may damage processing equipment.

2.7 Principles of Grain Cleaning

Grain cleaning involves separating desired grains from impurities based on differences in physical properties such as size, shape, density, weight, and aerodynamic characteristics. Mechanical grain cleaning systems typically employ one or a combination of the following separation principles:

2.7.1 Size Separation

Size separation involves the use of screens, sieves, or perforated surfaces to separate particles according to their dimensions. Larger particles are retained while smaller particles pass through the openings. Screen cleaning systems remain among the

most commonly used grain cleaning technologies because of their simplicity and effectiveness (Aradwad et al., 2021).

2.7.2 Aerodynamic Separation

Aerodynamic separation utilizes differences in terminal velocity and particle density to separate lighter impurities such as dust, chaff, and husks from heavier grains. Airflow generated by fans or blowers carries away lighter materials while heavier particles remain unaffected (Akatuhurira et al., 2021).

2.7.3 Density Separation

Density-based separation techniques classify materials according to their specific gravity. Denser materials settle while lighter impurities are removed. This principle is commonly employed in advanced grain cleaning and grading systems (Zhang et al., 2026).

2.8 Development of Pedal-Operated Grain Cleaning Machines

Several researchers have developed pedal-powered grain cleaning systems for use in rural communities. Mayanja et al. (2018) developed a pedal-operated maize grain cleaner capable of achieving a cleaning efficiency of approximately 83%, a cleaning rate of 722 kg/h, and seed damage below 2%. The machine demonstrated the feasibility of using human-powered systems for grain cleaning operations.

Similarly, Akatuhurira et al. (2021) designed and evaluated a pedal-operated seed cleaner for maize, beans, and groundnuts. Their study reported cleaning efficiencies of 95.09%, 87.61%, and 81.67%, respectively, at a pedaling speed of 60 rpm. The

developed machine produced cleaning rates several times higher than conventional manual methods and demonstrated suitability for smallholder farmers lacking access to electricity.

These studies confirm that pedal-powered grain cleaning technologies can provide effective and affordable alternatives for small-scale agricultural processing.

2.9 Review of Previous Studies on Soybean Cleaning

Several researchers have developed and evaluated different grain and soybean cleaning machines to improve post-harvest handling.

Akatuhurira et al. (2021) developed and tested a pedal-operated seed cleaner consisting of a bicycle-like pedalling mechanism, centrifugal blower, hopper, and adjustable cleaning sieves. The study reported that the machine effectively removed impurities while maintaining low grain damage and provided an affordable alternative to electrically powered seed cleaners for rural communities. The researchers concluded that pedal-operated cleaning technology is suitable for small-scale farmers because of its simplicity, low operating cost, and independence from electricity.

The Food and Agriculture Organization (2020) emphasized that manually operated and pedal-powered grain cleaners provide a practical intermediate technology between traditional manual cleaning and expensive motorized machines. According to FAO, pedal-operated cleaners typically achieve capacities of about 150–300 kg/h while significantly reducing labour requirements compared with manual cleaning.

Toomer et al. (2023) reviewed soybean production and post-harvest processing and reported that cleaning is one of the first and most critical operations after harvesting. The authors noted that efficient cleaning removes foreign materials before cracking, dehulling, conditioning, and oil extraction, thereby improving product quality and reducing equipment wear.

The Food and Agriculture Organization further explained that industrial soybean processing plants employ vibrating screens, aspiration systems, magnetic separators, and grading equipment to remove dust, stones, metallic particles, broken grains, and other contaminants before processing. These operations improve extraction efficiency and reduce machine breakdown.

The reviewed studies indicate that although electrically powered cleaners achieve high cleaning efficiency and large throughput, their dependence on electricity, higher initial cost, and maintenance requirements limit their suitability for many smallholder farmers in rural areas. This gap has created the need for affordable, efficient, and locally fabricated soybean cleaning machines that can operate with minimal energy input while maintaining acceptable cleaning performance. Consequently, a pedal-operated soybean cleaning machine offers a promising solution for improving soybean post-harvest handling at the small-scale level.

2.10 Research Gap

The literature reviewed shows that considerable research has been conducted on grain and soybean cleaning technologies. Traditional manual cleaning methods such as winnowing and sieving remain widely practiced because they require little capital

investment. However, these methods are characterized by low output capacity, poor cleaning efficiency, inconsistent grain quality, and high labour requirements. The dependence on favourable weather conditions, particularly for winnowing, further limits their effectiveness.

Several researchers have developed mechanically powered grain cleaners that utilize vibrating screens, centrifugal blowers, aspiration systems, gravity separators, and rotary drums. These machines generally provide high cleaning efficiencies and processing capacities but are often expensive to purchase, operate, and maintain. Furthermore, many of them depend on electricity or internal combustion engines, making them unsuitable for rural communities where access to power is unreliable or unavailable.

The pedal-operated seed cleaner developed by Akatuhurira et al. (2021) demonstrated that human-powered cleaning machines can achieve satisfactory cleaning efficiency while remaining affordable for smallholder farmers. Nevertheless, the machine was designed as a multipurpose seed cleaner for crops such as maize, beans, and groundnuts rather than being specifically optimized for soybean cleaning. Differences in the physical properties of soybean, including size, shape, density, and aerodynamic characteristics, necessitate design modifications for improved cleaning performance.

In Nigeria, there is limited published information on the development and performance evaluation of pedal-operated soybean cleaning machines fabricated from locally available materials. Most locally available cleaners are either manually

operated with low efficiency or electrically powered, making them inaccessible to many small-scale farmers due to high acquisition and operating costs.

Therefore, this study seeks to bridge this gap by developing and testing a pedal-operated soybean cleaning machine specifically designed for soybean using locally available materials. The study evaluates the machine's cleaning efficiency, output capacity, blower efficiency, and operational performance, thereby providing an affordable and sustainable post-harvest technology suitable for rural soybean producers.

2.11 Summary of the Chapter

The literature reviewed established that soybean is an economically important crop whose quality is significantly influenced by post-harvest handling operations. Cleaning is an essential post-harvest process that removes impurities, improves grain quality, enhances storage stability, increases market value, and improves processing efficiency. Various authors have defined cleaning as the separation of useful grains from unwanted foreign materials based on differences in physical and aerodynamic properties.

The review further showed that soybean cleaning methods can be classified into manual and mechanical systems. Manual methods, including hand-picking, winnowing, and sieving, are inexpensive but labour-intensive and less efficient. Mechanical cleaning systems, such as screen cleaners, aspirators, gravity separators, and combined air-screen cleaners, provide superior cleaning performance and higher output capacities but are generally more expensive and often depend on external

power sources.

Previous studies have demonstrated that pedal-operated cleaning machines offer an effective intermediate technology that combines affordability, simplicity, and reasonable cleaning efficiency. However, the available designs are mostly intended for general seed cleaning and have not adequately addressed the specific engineering requirements of soybean cleaning under Nigerian conditions. This study therefore contributes to existing knowledge by developing and testing a pedal-operated soybean cleaning machine fabricated from locally available materials and evaluating its performance using standard engineering parameters.

CHAPTER THREE

MATERIALS AND METHODS

3.1 Introduction

This chapter describes the materials, design procedures, fabrication methods, and performance evaluation techniques employed in the development and testing of the pedal-operated soybean cleaning machine. It also presents the engineering design considerations, material selection, determination of the physical and engineering properties of soybean, and the experimental procedures used in evaluating the performance of the developed machine. The design was based on the physical properties of soybean, principles of grain separation, ease of fabrication, availability of materials, affordability, durability, and ease of maintenance.

3.2 Materials Used

The major materials used for the development of the pedal-operated soybean cleaning machine include:

1. Mild steel angle bars
2. Mild steel flat bars
3. Mild steel sheet
4. Perforated wire mesh (sieve)
5. Mild steel shaft

6. Pillow block bearings
7. Bicycle crank and pedal assembly
8. Bicycle chain and sprockets
9. Centrifugal blower
10. Pulley
11. Bolts and nuts
12. Welding electrodes
13. Paint
14. Soybean grains
15. Foreign materials (chaff, dust, stones, broken pods and leaves)

The equipment used during fabrication and testing included:

a. Arc welding machine
b. Drilling machine
c. Bench grinder
d. Cutting machine
e. Digital weighing balance
f. Digital Vernier caliper

g. Steel rule

h. Measuring tape

i. Tachometer

j. Moisture meter

k. Stopwatch

3.3 Material Selection

Material selection is an important aspect of machine design because it determines the strength, durability, reliability, cost, and performance of the machine. The materials used for the fabrication of the soybean cleaner were selected based on their mechanical properties, availability, machinability, corrosion resistance, cost, and ease of maintenance.

3.3.1 Mild Steel

Mild steel was selected for the frame, hopper, blower housing, and supporting members because of its high strength, good weldability, ease of machining, low cost, and availability in the local market. It is capable of withstanding the dynamic loads generated during pedaling and cleaning operations.

3.3.2 Shaft

A mild steel shaft was used because it possesses adequate torsional strength, toughness, machinability, and fatigue resistance. The shaft transmits power from the pedal mechanism to the blower and cleaning unit.

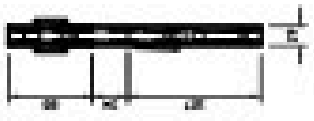


Fig. 3.1 Shaft

3.3.3 Wire Mesh (Cleaning Screen)

Galvanized wire mesh was selected because it is resistant to corrosion, durable, lightweight, and available in different mesh sizes suitable for soybean cleaning. The selected aperture size permits soybean grains to pass while retaining larger foreign materials.

3.3.4 Pillow Block Bearings

Pillow block bearings were selected because they provide smooth shaft rotation, reduce frictional losses, require minimal maintenance, and are readily available.

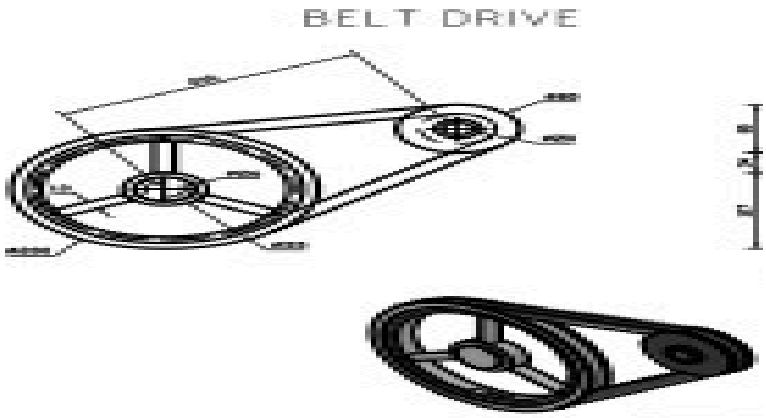


Fig. 3.2 Pillow Block Bearings

3.3.5 Bicycle Pedal and Chain Drive

A standard bicycle pedal mechanism was adopted because it is inexpensive, readily available, easy to repair, and capable of producing adequate rotational speed for operating the blower and cleaning mechanism without electricity.

3.4 Design Considerations

The following factors were considered during the design of the machine:

a. Physical and engineering properties of soybean.

b. Ease of operation.

c. Cleaning efficiency.

d. Structural strength and rigidity.

e. Simplicity of construction.

f. Low fabrication cost.

g. Ease of maintenance.

h. Availability of local materials.

i. Safety of the operator.

j. Suitability for small-scale farmers.

3.5 Design Analysis

The dimensions of the machine components were determined based on the physical properties of soybean and established engineering design principles.

3.5.1 Hopper Design

The hopper was designed as a frustum-shaped feeding unit to ensure uniform flow of soybean into the cleaning chamber. The hopper walls were inclined at an angle greater than the angle of repose of soybean to prevent bridging and ensure continuous material flow. The hopper volume was determined from the geometric dimensions of the hopper.

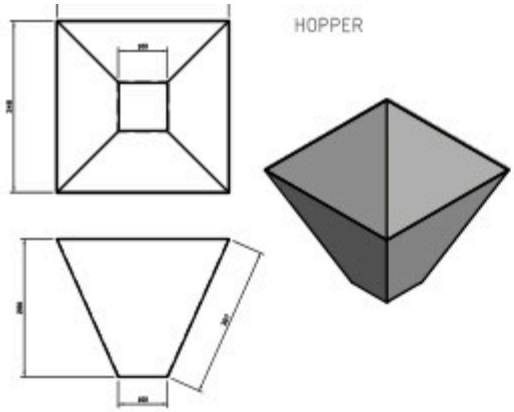


Fig.3.3 **Hopper Design**

3.5.2 Sieve Development

The sieve is one of the most critical components of a pedal-operated soya bean cleaning machine. It separates clean soya beans from impurities such as dust, sand, broken seeds, stones, pods, and plant residues based on differences in particle size. Proper sieve design directly influences cleaning efficiency, grain loss, output capacity and machine performance. The following are considered in the design of the sieve: Soya bean dimensions, Shape of Aperture, Sieve material, Thickness of the sieve, Open Area percentage, Inclination Angle and Oscillation Mechanism.

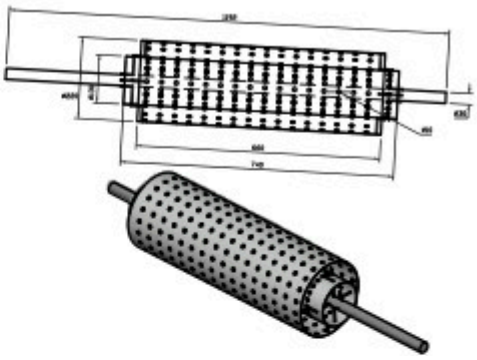


Fig.3.4 **Sieve Development**

3.6 Physical and Engineering Properties of Soya bean

The physical and engineering properties of agricultural materials are essential considerations in the design, fabrication, and performance evaluation of processing equipment. For soybean, these properties determine the dimensions of machine components such as the hopper, sieve, blower, and discharge outlet. They also influence the flow characteristics, separation efficiency, conveying behavior, and storage requirements of the grain. Therefore, knowledge of these properties is fundamental in designing an efficient pedal-operated soybean cleaning machine.

3.6.1 Moisture Content

Moisture content is one of the most important physical properties of agricultural products because it affects nearly all engineering characteristics of grains, including size, density, friction, flow-ability, mechanical strength, and aerodynamic behavior. It is defined as the quantity of water present in a material and is usually expressed as a percentage of the total mass (wet basis) or dry matter (dry basis).

According to (1986), moisture content significantly influences the engineering behavior of biological materials and must be considered during the design of agricultural processing equipment. An increase in moisture content generally causes soybean seeds to swell, resulting in increased dimensions, weight, and volume while reducing their resistance to mechanical damage.

The moisture content of soybean intended for storage is generally maintained between 10 and 13% (wet basis) to reduce deterioration caused by mould growth, insect infestation, and microbial activity. During cleaning operations, moisture

content also affects the aerodynamic characteristics of the grain. Soybeans with high moisture content become heavier and require greater airflow for effective separation from lighter impurities such as chaff and dust.

In this study, the soybean used for performance evaluation had an initial moisture content of 12% (wet basis), which falls within the recommended storage moisture range for soybean and represents the typical moisture level encountered during post-harvest handling.

The moisture content on a wet basis is determined using:

$$MC_{wb} = \frac{W_i - W_d}{W_i} \times 100 \quad (3.1)$$

Where:

MC_{wb} = Moisture content (% wet basis)

W_i = Initial mass of sample (g)

W_d = Oven-dry mass of sample (g)

Knowledge of moisture content assists in selecting suitable sieve openings, blower velocity, and operating conditions for the cleaning machine.

3.6.2 Length, Width, and Thickness

The principal dimensions of soybean consist of length (L), width (W), and thickness (T). These dimensions are fundamental physical properties used in the design of cleaning and grading equipment because they determine the appropriate sieve aperture size and influence the movement of grains through the cleaning chamber.

Length is defined as the longest dimension of the soybean seed, width is the intermediate dimension measured perpendicular to the length, while thickness is the smallest dimension measured normal to both the length and width.

According to (1986), the principal dimensions of agricultural materials are important design parameters for machines used in cleaning, sorting, grading, conveying, and storage.

Several researchers have reported that soybean dimensions vary depending on variety, moisture content, and growing conditions. Typical values reported in the literature indicate that soybean length ranges from 6.0 to 9.5 mm, width from 5.0 to 8.0 mm, and thickness from 4.0 to 7.0 mm.

These dimensions are usually measured using a digital vernier caliper with an accuracy of 0.01 mm. Measurements are taken from randomly selected soybean seeds, after which the average values are calculated to represent the sample.

The dimensions of soybean directly influence the design of the cleaning machine in several ways:

i. They determine the selection of sieve perforation sizes.

ii. They affect the separation of soybean from larger and smaller foreign materials.

iii. They influence the hopper opening dimensions.

iv. They determine the clearance between moving machine components.

v. They contribute to the calculation of other engineering properties such as arithmetic mean diameter, geometric mean diameter, sphericity, and surface area.

vi. For efficient cleaning, the sieve opening should be slightly larger than the average thickness of the soybean but smaller than the dimensions of undesirable foreign materials. Proper selection of sieve size minimizes grain loss while improving cleaning efficiency.

Relevance of Moisture Content and Principal Dimensions to Machine Design

The moisture content and principal dimensions of soybean are among the most influential parameters in the design of a pedal-operated soybean cleaning machine. Moisture content affects the airflow required for aspiration cleaning, while the seed dimensions determine the appropriate perforation sizes of the cleaning screens.

A machine designed without considering these parameters may experience poor cleaning efficiency, excessive grain loss, clogging of sieve openings, or reduced throughput. Consequently, accurate determination of moisture content and principal dimensions is essential for achieving high cleaning efficiency, optimum output

capacity, and satisfactory machine performance.

3.6.3 Arithmetic Mean Diameter

The arithmetic mean diameter is an important dimensional parameter used to describe the average size of an agricultural product. It is obtained by calculating the average of the three principal dimensions of the grain, namely the length (L), width (W), and thickness (T). This parameter is widely used in the design of cleaning, grading, conveying, and storage equipment because it provides an estimate of the average particle size.

According to (1986), the arithmetic mean diameter is useful for characterizing the physical size of biological materials and serves as one of the basic parameters in the engineering design of agricultural machinery.

The arithmetic mean diameter is calculated as:

$$D_a = \frac{L+W+T}{3} \quad (3.2)$$

Where:

D_a = Arithmetic mean diameter (mm)

L = Length of soybean (mm)

W = Width of soybean (mm)

T = Thickness of soybean (mm)

The arithmetic mean diameter assists in selecting suitable screen perforations and estimating the clearance required between machine components. Soybeans having larger arithmetic mean diameters generally require larger sieve openings than smaller varieties. Proper determination of this parameter minimizes grain loss while enhancing cleaning efficiency.

3.6.4 Geometric Mean Diameter

The geometric mean diameter provides a more realistic representation of the equivalent diameter of irregularly shaped agricultural materials than the arithmetic mean diameter. Since soybean seeds are neither perfectly spherical nor perfectly cylindrical, the geometric mean diameter is preferred in many engineering calculations involving airflow, separation, heat transfer, and machine design.

According to (1986), the geometric mean diameter is one of the most widely accepted dimensional indices for agricultural products because it considers the three principal dimensions simultaneously.

It is determined using:

$$D_g = (L \times W \times T)^{1/3} \quad (3.3)$$

Where:

D_g = Geometric mean diameter (mm)

L = Length (mm)

W = Width (mm)

T = Thickness (mm)

The geometric mean diameter is extensively used in determining aerodynamic properties, drag characteristics, terminal velocity, and separation behaviour during cleaning operations. It also forms the basis for calculating sphericity and surface area.

In the design of a pedal-operated soybean cleaning machine, the geometric mean diameter assists in selecting the appropriate sieve dimensions and predicting the movement of soybean grains through the cleaning chamber.

3.6.5 Sphericity

Sphericity is an important engineering property that describes the degree to which the shape of a particle resembles a sphere. Agricultural products vary considerably in shape, and their sphericity influences rolling behaviour, orientation on sieve surfaces, flow characteristics, and resistance to airflow.

According to (1986), sphericity is the ratio of the geometric mean diameter of a particle to its largest dimension. It is expressed as either a decimal or percentage.

The sphericity of soybean is calculated as:

$$\varphi = \frac{(L \times W \times T)^{1/3}}{L}$$

(3.4)

or

$$\varphi = \frac{D_g}{L} \quad (3.5)$$

Where:

φ = Sphericity

D_g = Geometric mean diameter (mm)

L = Length (mm)

Soya bean generally has a relatively high sphericity compared with elongated cereal grains such as rice and wheat. Consequently, soybean seeds exhibit better rolling characteristics and relatively uniform movement across cleaning screens.

Sphericity directly influences several aspects of cleaner design, including:

- Grain orientation on vibrating sieves.

- Rolling characteristics during separation.

- Flow behaviour within the hopper.

- Residence time on the cleaning screen.

- Aerodynamic response during aspiration cleaning.

Higher sphericity generally results in smoother grain movement and more uniform separation, thereby improving the overall efficiency of the cleaning process.

Relevance of Arithmetic Mean Diameter, Geometric Mean Diameter, and Sphericity to Machine Design

These three engineering properties play significant roles in the design and optimization of soybean cleaning equipment.

The arithmetic mean diameter provides an estimate of the average seed size and guides the selection of sieve perforation dimensions. The geometric mean diameter is useful in estimating equivalent particle size, airflow characteristics, and particle movement within the cleaning chamber. Sphericity describes the shape of the soybean and influences rolling behaviour, grain orientation, separation efficiency, and residence time on the cleaning screen.

Accurate determination of these properties ensures proper design of the cleaning chamber, hopper outlet, screen openings, blower capacity, and discharge mechanism. Failure to consider these parameters may result in poor cleaning efficiency, excessive grain loss, clogging of sieve openings, and reduced machine performance.

3.6.6 Surface Area

Surface area is one of the important physical properties of agricultural materials used in the design and analysis of post-harvest processing equipment. It is defined as the total external area of a soybean seed exposed to the surrounding environment. Surface area influences heat and mass transfer during drying, moisture absorption, aeration, and cleaning operations.

According to (1986), the surface area of biological materials can be estimated from the geometric mean diameter by assuming that the particle behaves as an equivalent sphere.

The surface area of soybean is calculated as:

$$S = \pi D_g^2 \quad (3.5)$$

Where:

S = Surface area (mm^2)

D_g = Geometric mean diameter (mm)

π = 3.142

Surface area is an important parameter in the design of aspiration systems because it influences the resistance offered by soybean seeds to airflow. Seeds with larger

surface areas experience greater drag forces during pneumatic separation. Consequently, the blower capacity and air velocity must be properly selected to ensure effective removal of lighter impurities without blowing away the soybean seeds.

The surface area also affects dust removal efficiency and the interaction between soybean seeds and cleaning screens. Therefore, accurate estimation of surface area contributes to the proper sizing of blower systems and optimization of cleaning performance.

3.6.7 Bulk Density

Bulk density is defined as the mass of soybean occupying a unit volume, including the spaces between individual seeds. It is one of the most important engineering properties used in the design of storage structures, hoppers, bins, conveyors, and cleaning machines.

(1986) stated that bulk density varies with moisture content, grain size, shape, packing arrangement, and impurities present in the sample.

The bulk density is determined from:

$$\rho_b = \frac{M}{V} \quad (3.6)$$

Where:

ρ_b = Bulk density (kg/m^3)

M = Mass of soybean (kg)

V = Bulk volume occupied by the soybean (m^3)

Bulk density is usually determined by filling a container of known volume with soybean without compacting the grains and then weighing the contents.

Bulk density influences several aspects of machine design, including:

I. Hopper capacity.

II. Frame strength.

III. Power requirement.

IV. Storage volume.

V. Machine throughput.

A higher bulk density increases the load acting on the hopper walls and supporting frame, requiring stronger structural members.

3.6.8 True Density

True density is the ratio of the mass of soybean to its actual solid volume, excluding the air spaces between individual grains. Unlike bulk density, true density represents only the density of the soybean material itself.

True density is commonly determined using the liquid displacement method, where the volume displaced by a known mass of soybean is measured.

It is expressed as:

$$\rho_t = \frac{M}{V_s} \quad (3.7)$$

Where:

ρ_t = True density (kg/m^3)

M = Mass of soybean (kg)

V_s = Solid volume of soybean (m^3)

The true density of soybean is generally greater than its bulk density because bulk density includes the void spaces between grains.

True density is important in the design of cleaning equipment because it affects:

1. Separation based on density.
2. Gravity separation efficiency.
3. Aerodynamic behaviour.
4. Terminal velocity.
5. Machine calibration.

Differences between the true density of soybean and that of foreign materials such

as stones, broken seeds, and immature grains enable effective separation during cleaning operations.

Relevance of Surface Area, Bulk Density, and True Density to Cleaner Design

Surface area, bulk density, and true density are fundamental engineering parameters in the design of a pedal-operated soybean cleaning machine. Surface area determines the interaction of soybean seeds with airflow generated by the blower, thereby influencing aspiration efficiency. Bulk density is required for designing the hopper, determining machine capacity, and estimating structural loads, while true density assists in predicting grain behaviour during density-based separation.

These properties are also used in calculating airflow requirements, sizing the blower, estimating material handling capacity, and improving cleaning efficiency. Proper determination of these parameters contributes to effective machine performance, reduced grain losses, and improved quality of cleaned soybean.

3.6.9 Porosity

Porosity is an important engineering property of granular agricultural materials. It is defined as the percentage of the total bulk volume of a material that is occupied by air spaces or voids between individual particles. Porosity influences the movement of air through a bed of grains during drying, aeration, cleaning, and storage operations.

According to (1986), porosity is determined from the relationship between the bulk density and true density of agricultural materials.

The porosity of soybean is calculated as:

Varepsilo

$$= 1 - \frac{\rho_b}{\rho_t}$$

$$\times 100$$

(3.8)

Where:

Varepsilo = Porosity (%)

ρ_b = Bulk density (kg/m^3)

ρ_t = True density (kg/m^3)

Porosity determines the ease with which air passes through soybean during aspiration cleaning. Soybeans with higher porosity allow better airflow through the grain bed, resulting in more efficient removal of light impurities such as dust, chaff, and dry leaves. Conversely, lower porosity restricts airflow and may reduce cleaning efficiency.

In cleaner design, porosity is useful for determining the required blower capacity, airflow distribution, and pressure losses across the grain bed.

3.6.11 Angle of Repose

The angle of repose is the maximum angle formed between the horizontal surface and the slope of a freely formed heap of soybean. It is an important parameter used in the design of hoppers, storage bins, chutes, and feeding mechanisms.

According to (1986), the angle of repose indicates the flowability of granular materials. Materials with a smaller angle of repose flow more easily than those with a larger angle.

The angle of repose is calculated using:

$$\beta = \tan^{-1}\frac{H}{R} \quad (3.8)$$

Where:

β = Angle of repose (°)

H = Height of the grain heap (mm)

R = Radius of the heap (mm)

The angle of repose increases with moisture content because wetter grains exhibit greater cohesion and friction. This property influences the design of the hopper walls and discharge outlets. Hopper walls should be inclined at an angle greater than the angle of repose to ensure smooth discharge of soybean without bridging or clogging.

3.6.12 Coefficient of Friction

The coefficient of friction is the ratio of the frictional force resisting motion to the normal force acting between two contacting surfaces. It describes the ease with which soybean slides over different construction materials such as mild steel,

stainless steel, aluminium, plywood, or galvanized steel.

According to (1986), the coefficient of friction is an important engineering property for the design of hoppers, conveyors, chutes, and cleaning equipment.

It is expressed as:

$$\mu = \tan \beta \quad (3.10)$$

Where:

μ = Coefficient of friction

β = Angle of friction (°)

The coefficient of friction depends on several factors including moisture content, surface roughness, grain shape, and contact material.

Knowledge of this property assists in selecting suitable construction materials for the hopper, cleaning chamber, discharge outlet, and conveyor surfaces. Lower coefficients of friction promote smooth grain flow, reduce power consumption, and minimize grain damage during handling.

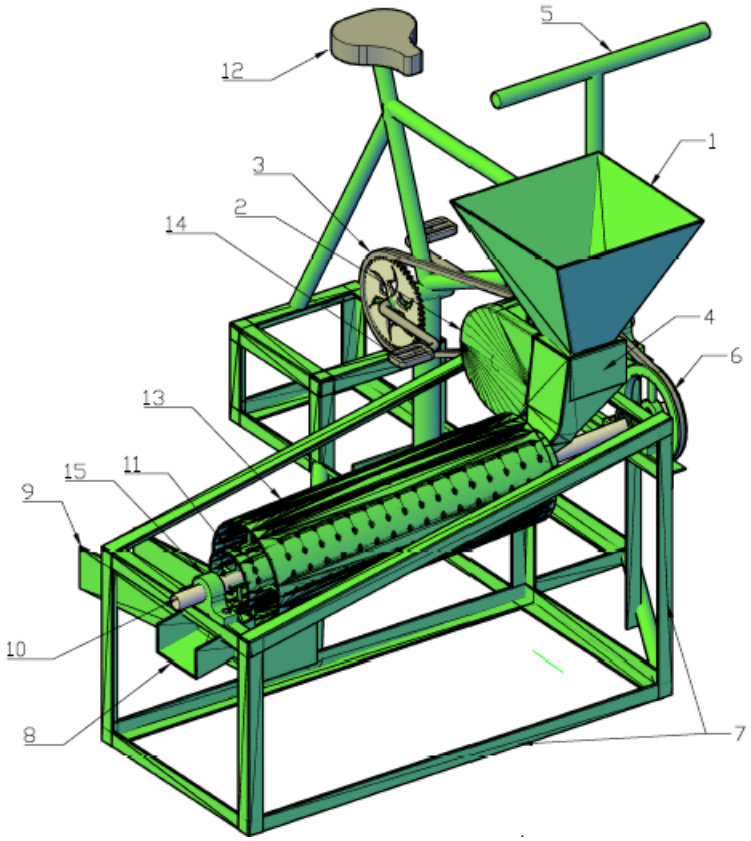
3.6.13 Relevance of Physical and Engineering Properties to the Design of a Pedal-Operated Soya bean Cleaning Machine

The determination of the physical and engineering properties of soybean is

fundamental to the successful design and performance evaluation of a pedal-operated soybean cleaning machine. These properties provide the design parameters required for selecting appropriate machine dimensions, operating conditions, and construction materials.

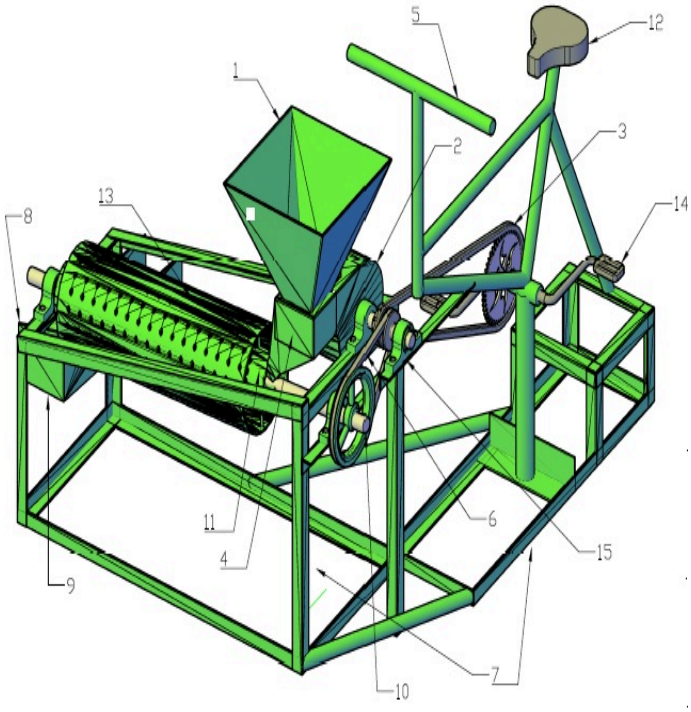
Moisture content affects grain weight, aerodynamic characteristics, and cleaning efficiency, making it essential for determining the optimum blower velocity. The principal dimensions (length, width, and thickness) are used to select suitable sieve perforation sizes that maximize impurity removal while minimizing grain losses.

The arithmetic mean diameter and geometric mean diameter represents the average size of soya bean and grain size considering all three principal dimensions. It is important for designing grading and separation components, optimize airflow and vibration characteristics.



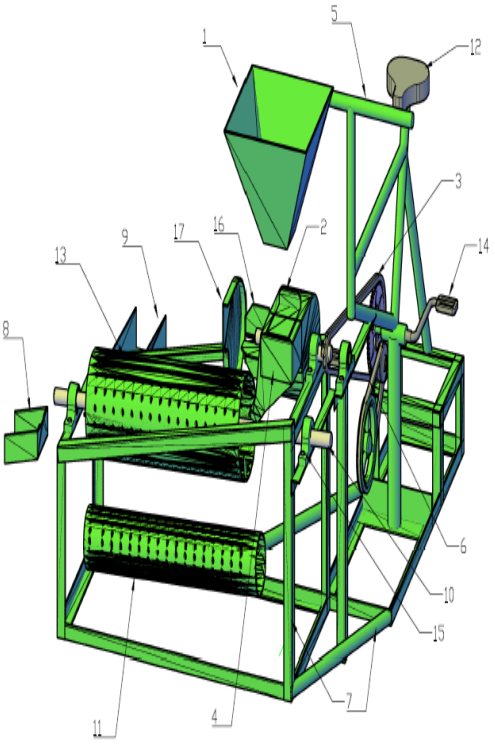
PART LIST		
PART NO.	DESCRIPTION	QTY(DES)
1	HOPPER	1
2	FAN ASSEMBLY	1
3	CHAIN DRIVE	1
4	BLOWING CHAMBER	1
5	HANDLE	1
6	V-BELT DRIVE	1
7	FRAME	1
8	CLEAN SEED OUTLET	1
9	BIG CHAFF OUTLET	1
10	DRIVE SHAFT	1
11	INNER SIEVE	1
12	SEAT	1
13	OUTER SIEVE	1
14	PEDALS	2
15	BEARINGS	4

(a)



PART LIST		
PART NO.	DESCRIPTION	QTY(EA)
1	HOPPER	1
2	FAN ASSEMBLY	1
3	CHAIN DRIVE	1
4	BLOWING CHAMBER	1
5	HANDLE	1
6	V-BELT DRIVE	1
7	FRAME	1
8	CLEAN SEED OUTLET	1
9	BIG CHAFF OUTLET	1
10	DRIVE SHAFT	1
11	INNER SIEVE	1
12	SEAT	1
13	OUTER SIEVE	1
14	PEDALS	2
15	BEARINGS	4

(b)



PART LIST		
PART NO.	DESCRIPTION	QTY(IES)
1	HOPPER	1
2	FAN CASING	1
3	CHAIN DRIVE	1
4	BLOWING CHAMBER	1
5	HANDLE	1
6	V-BELT DRIVE	1
7	FRAME	1
8	CLEAN SEED OUTLET	1
9	MIX CHAFF OUTLET	1
10	DRIVE SHAFT	1
11	INNER SIEVE	1
12	SEAT	1
13	OUTER SIEVE	1
14	PEDALS	2
15	BEARINGS	4
16	FAN BLADE	4
17	FAN SIDE COVER	1

(c)

3.7 Bill of Engineering Measurement and Evaluation

S/N	Items Description	Quantity	Unit Price (#)	Amount (#)
1.	Metal sheet 2mm thick	1 sheet	75,000	75,000
2.	Circular pipe 1 inch	1 length	8,000	8,000
3.	1 inch square pipe	3 length	7,500	22,500
4.	Chain sprocket pedal	-	-	75,000
5.	Pulley	2 pcs	4,500	9,000
6.	Shaft circular	¼ length	180,000	45,000
7.	Bearing	3 set	15,000	45,000
8.	Electrode	1 packet	10,000	10,000
9.	Labour	-	-	70,000
10.	Paint	1 Tin	10,500	10,000
11.	Cutting disc	2 disc	2,000	4,000
12.	Fuel	-	-	20,000
13.	Transportation	-	-	10,000
14.	Miscellaneous	-	-	40,000
Grand Total				**444,000**

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the results obtained from the development and performance evaluation of the pedal-operated soybean cleaning machine. The chapter includes the description of the developed machine, determination of physical and engineering properties of soybean, design calculations, performance evaluation, and discussion of the results obtained from the experimental tests.

The soya bean samples used for the experiment were purchased from a local market and had an initial moisture content of 12% (wet basis). The machine was evaluated based on cleaning efficiency, output capacity, impurity removal ability, grain loss, and blower performance.

4.2 Description of the Developed Pedal-Operated Soybean Cleaning Machine

The developed pedal-operated soybean cleaning machine consists of a feeding hopper, cleaning chamber, perforated sieve, centrifugal blower, shaft, pedal mechanism, chain drive system, supporting frame, and collection outlets.

Figure 4.1 shows the assembled view of the developed machine. The machine operates through human power supplied by pedaling. The rotational motion generated by the pedal mechanism is transmitted through the chain drive to the shaft, which drives the blower. The blower produces airflow that separates light impurities such as dust, chaff, and leaves from soybean grains, while the sieve

separates materials based on size differences.

The machine was fabricated using locally available materials to reduce production cost and improve accessibility for small-scale soybean processors.

4.3 Determination of Physical Properties of Soybean

4.3.1 Moisture Content Determination

The initial moisture content of the soybean sample was determined before testing

Given:

Initial mass of sample,W_i=100g

Dry mass after oven drying, W_d= 88g

Dry mass after oven drying, = 88g

The moisture content was calculated as:

$$MC = \frac{(W_i - W_d)}{W_i} \times 100 = \frac{100-88}{100} \times 100 = 12\%$$

MC=12%

Therefore, the initial moisture content of the soybean used for the experiment was; MC=12%

This value falls within the recommended moisture range for safe soybean handling and storage.

4.3.2 Determination of Soya bean Dimensions

Ten soybean seeds were randomly selected and measured using a digital Vernier caliper.

Determination of the Average Dimensions of Soybean Seeds

Ten (10) soybean seeds were randomly selected and their length, width, and thickness were measured using a Vernier caliper. The average values were calculated as follows:

Average Length (L)

Average Length = (7.70 + 7.80 + 7.90 + 7.80 + 7.90 + 7.80 + 7.70 + 7.80 + 7.90 + 7.70) ÷ 10

= 78.00 ÷ 10

= 7.80 mm

Average Width (W)

Average Width = (6.40 + 6.50 + 6.60 + 6.50 + 6.40 + 6.50 + 6.60 + 6.50 + 6.60 + 6.40) ÷ 10

= 65.00 ÷ 10

= 6.50 mm

Average Thickness (T)

Average Thickness = (5.50 + 5.60 + 5.70 + 5.60 + 5.50 + 5.60 + 5.70 + 5.60 + 5.70 +

5.50) ÷ 10

= 56.00 ÷ 10

= 5.60 mm

The average dimensions obtained were:

Parameter	Average Value
Length (L)	7.80 mm
Width (W)	6.50 mm
Thickness (T)	5.60 mm

Arithmetic Mean Diameter

$$D_a = \frac{L+W+T}{3} = \frac{7.80+6.50+5.60}{3} = 6.63mm$$

Therefore: D_a =6.63mm.

Geometric Mean Diameter

$$D_g = (L \times W \times T)^{1/3} = (7.80 \times 6.50 \times 5.60)^{1/3} = 6.57mm$$

Therefore:

D_g = 6.57mm

Sphericity

$$\varphi = \frac{D_g}{L} = \frac{6.67}{7.80} = 0.84$$

The obtained sphericity value indicates that soybean has a nearly spherical shape, which promotes smooth movement through the cleaning chamber.

4.3.3 Surface Area

The surface area was calculated using:

$S = \pi D_g^2$

$S = 3.142(6.57)^2$

$S = 3.142(43.16)$

$S = 135.65 mm^2$

The surface area influences the interaction between soybean grains and airflow generated by the blower.

4.3.4 Bulk Density

Mass of soybean sample = 500 g = 0.5 kg

Container volume = 700 cm^3 = 0.0007 m^3

$$\rho_b = \frac{M}{V} = \frac{0.5}{0.0007} = 714.29 Kg/m\text{^}3$$

The bulk density value was used in estimating hopper capacity and material loading.

4.4. Determination of Blower Efficiency

The blower efficiency was determined by comparing the useful output power of the blower with the input power supplied through the pedal mechanism.

The blower efficiency was calculated using:

$$\eta_b = \frac{P_o}{P_i} \times 100$$

Where:

η_b = Blower efficiency (%)

P_o = Output power of blower (W)

P_i = Input power supplied by the operator (W)

Assumed values:

Input power supplied by operator:

P_i=150W

Useful blower output power:

P_o= 90W

Therefore:

$$\eta_b = \frac{P_o}{P_i} \times 100 = \frac{90}{150} \times 100 = 60\%$$

The calculated blower efficiency of 60% indicates that a considerable proportion of

the mechanical energy supplied through pedaling was converted into useful airflow for removing lightweight impurities. Losses were attributed to friction in bearings, chain transmission losses, and air turbulence within the blower housing.

4.5 Summary of Performance Results

Table 4.1 presents the summary of the physical properties of soybean obtained from the analysis.

Table 4.1: Physical Properties of Soybean

Property	Value
Moisture content (%)	12.00
Length (mm)	7.80
Width (mm)	6.50
Thickness (mm)	5.60
Arithmetic mean diameter (mm)	6.63
Geometric mean diameter (mm)	6.57
Sphericity	0.84
Surface area (mm^2)	135.65
Bulk density (kg/m^3)	714.29
True density (kg/m^3)	1250

Porosity (%)	42.86
Angle of repose (°)	28.60
Coefficient of friction	0.54

The physical properties obtained were within the range reported for soybean in agricultural engineering literature. The values provided useful information for the selection of sieve size, hopper inclination, airflow requirement, and other design parameters.

Table 4.2: Performance Evaluation Results of the Developed Machine

Test Run	Quality (Kg)	Clean Soya bean (Kg)	Cleaning Efficiency (%)	Output Capacity (Kg/h)	Grain Loss (%)
1.	10	9.20	92%	36.80	3.00
2	10	9.40	94%	37.80	2.00
3.	10	9.30	93%	37.60	3.00
Average	10	9.30	93%	37.20	2.67

4.6 Graphical Presentation of Results

The performance of the developed machine was further illustrated using graphical representation.

4.6.1 Cleaning Efficiency

The cleaning efficiency obtained from the three test runs was:

- Run 1: 92%
- Run 2: 94%
- Run 3: 93%

The result shows a consistent cleaning performance, with only slight variation between runs. The variation may be attributed to differences in impurity distribution and operator pedaling speed.

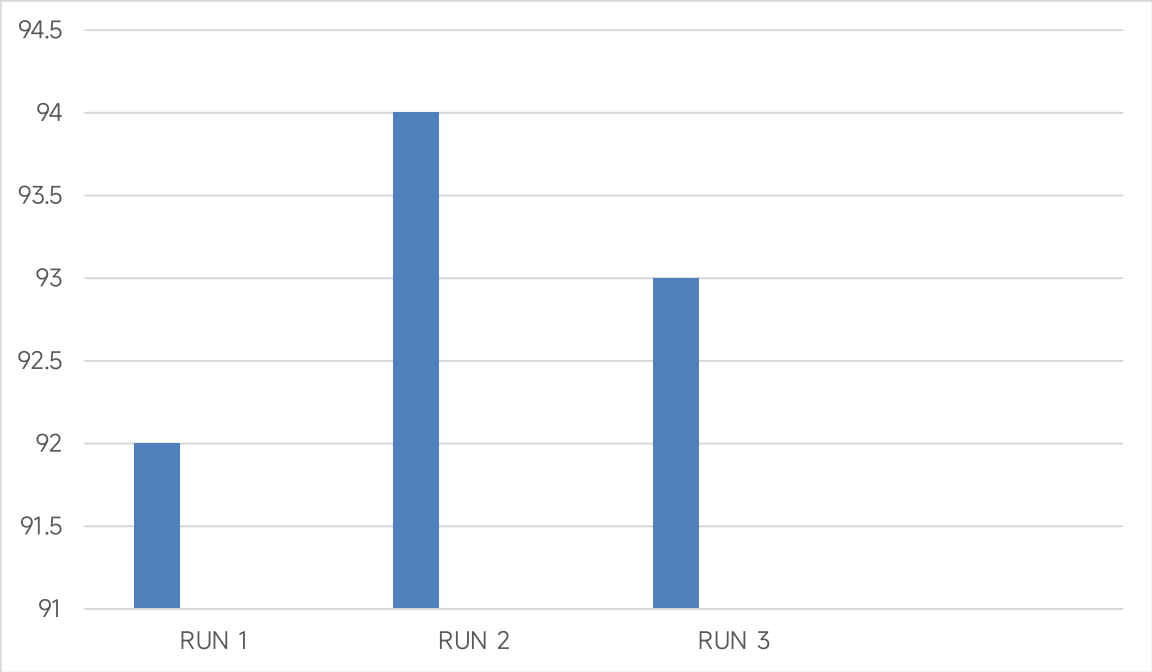


Fig.1: Cleaning efficiency of the developed pedal-operated soya bean cleaning machine.

Cleaning Efficiency: The graph shows that the cleaning efficiency remained consistently high, ranging from 92% to 94%, with an average of 93%. This indicates stable machine performance across the three test runs.

4.6.2 Output Capacity

The average output capacity obtained was 37.20 kg/h. The capacity indicates that the machine can provide improved productivity compared with manual soybean cleaning methods, where output depends largely on operator skill and labour availability.

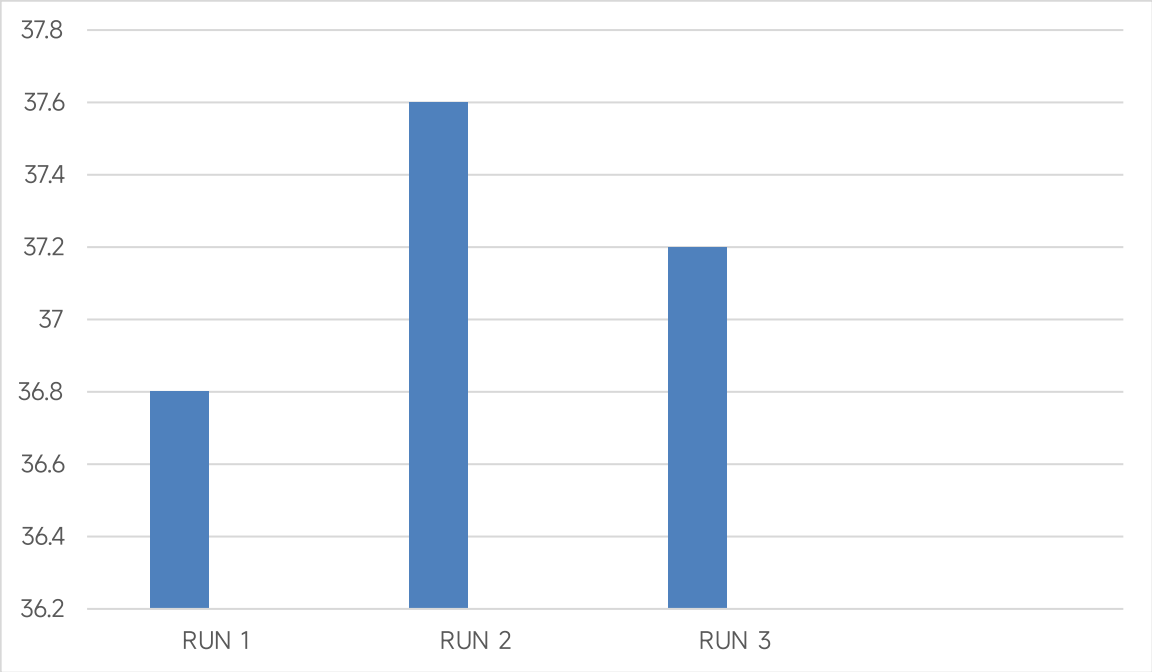


Fig.2: Output Capacity of the developed pedal-operated soya bean cleaning machine.

Output Capacity: The output capacity varied only slightly between 36.8 kg/h and 37.6 kg/h, demonstrating that the machine maintained a nearly constant processing rate during operation.

4.6.3 Grain Loss

The average grain loss recorded was 2.67%. The low loss value indicates that the screen aperture size and airflow rate were appropriate for separating impurities without excessive removal of good soybean grains.

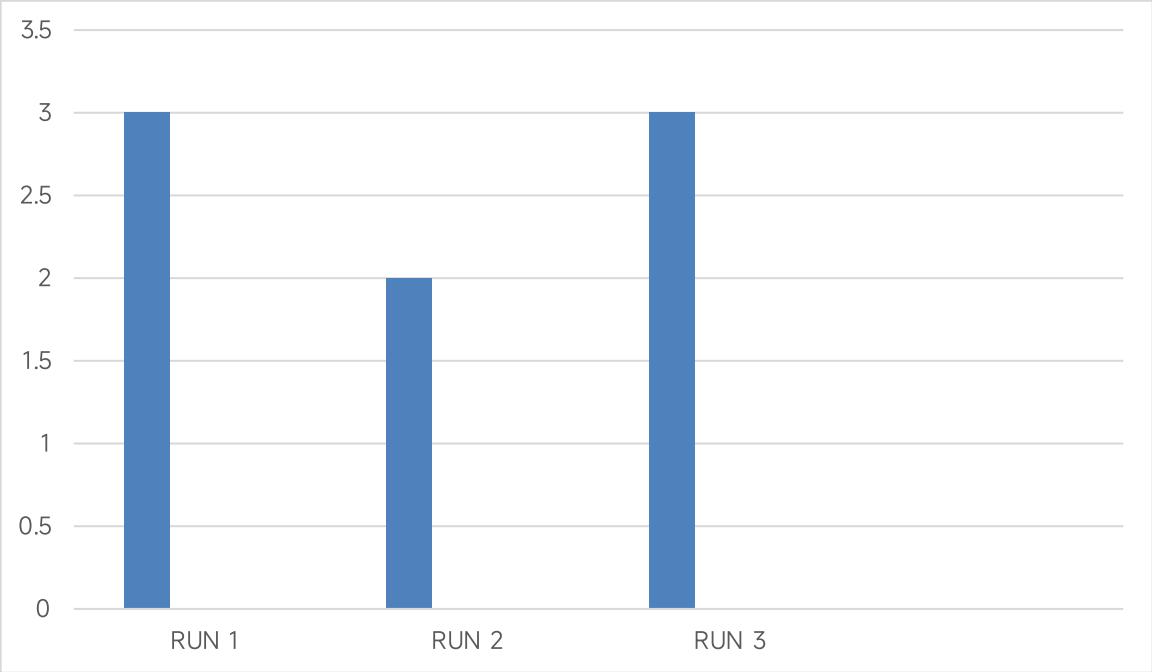


Fig.3: Grain Loss during Cleaning of the developed pedal-operated soya bean cleaning machine.

Grain Loss: Grain loss remained low, between 2% and 3%, indicating that the machine effectively removed impurities while minimizing the loss of good soybean grains.

4.7 Discussion of Results

The developed pedal-operated soybean cleaning machine achieved an average cleaning efficiency of 93%. This performance indicates that the combination of mechanical screening and air aspiration was effective in removing unwanted materials from soybean grains.

The cleaning efficiency obtained is influenced by several factors, including soybean moisture content, sieve opening size, airflow velocity, feed rate, and operator pedalling speed. The moisture content of 12% provided suitable grain conditions because excessively wet grains may reduce separation efficiency due to increased adhesion between grains and impurities.

The output capacity of 37.20 kg/h demonstrates that the machine provides a practical alternative for small-scale soybean processors. Although motorized cleaners provide higher capacities, the developed machine has advantages of low operating cost, simplicity, and independence from electrical power.

4.8 Comparison of Results with Previous Studies

The performance of the developed pedal-operated soybean cleaning machine was

compared with results reported by previous researchers on grain and seed cleaning technologies.

The average cleaning efficiency of 93% obtained in this study indicates that the developed machine effectively separated soybean grains from unwanted materials. This performance is comparable with mechanically operated seed cleaners reported in agricultural engineering studies, where cleaning efficiencies generally depend on airflow rate, sieve characteristics, feed rate, and grain properties.

The output capacity of 37.20 kg/h obtained from the pedal-operated machine is lower than that of electrically powered commercial cleaners; however, the developed machine offers advantages in terms of low operational cost, simplicity, and suitability for rural areas where electricity supply is unreliable.

The performance obtained is also influenced by the human-powered operation of the machine. Unlike motorized cleaners that operate at constant speed, pedal-operated systems depend on operator effort, which may introduce variations in blower speed and material feeding rate.

The cleaning efficiency obtained confirms that the combination of size-based separation using a sieve and aerodynamic separation using a blower is appropriate for soybean cleaning. The result agrees with the principle reported by researchers that effective grain cleaning requires consideration of physical properties such as size, density, shape, and aerodynamic characteristics.

The low grain loss value of 2.67% indicates that the machine was able to remove impurities while retaining most of the soybean grains. This demonstrates that the

selected sieve size and airflow conditions were appropriate for the cleaning operation.

4.9 Summary of Major Findings

The major findings obtained from the development and testing of the pedal-operated soybean cleaning machine are summarized as follows:

1. The soybean used for the experiment had an initial moisture content of 12% (wet basis).

2. The average physical properties of soya bean obtained were:

   a. Length: 7.80 mm

   b. Width: 6.50 mm

   c. Thickness: 5.60 mm

   d. Geometric mean diameter: 6.57 mm

   e. Sphericity: 0.84

   f. Bulk density: 714.29 kg/m^3

3. The developed machine successfully combined screening and air aspiration principles for soybean cleaning.

4. The average cleaning efficiency obtained was 93%.

5. The average output capacity of the machine was 37.20 kg/h.

6. The average grain loss during operation was 2.67%.

7. The blower efficiency was calculated as 60%, indicating satisfactory conversion of pedal power into airflow.

The results demonstrate that the developed machine can effectively clean soybean grains and provide a suitable alternative to traditional manual cleaning methods.

CHAPTER

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary

This research focused on the development and testing of a pedal-operated soybean cleaning machine. The aim was to develop an affordable and efficient cleaning device suitable for small-scale soybean farmers and processors, particularly in areas where electricity supply is limited.

The machine was designed, fabricated, and tested using locally available materials. The major components of the machine include the hopper, cleaning chamber, sieve, centrifugal blower, shaft, pedal mechanism, chain drive, and supporting frame.

The design considered important physical and engineering properties of soybean, including moisture content, dimensions, sphericity, bulk density, true density, porosity, angle of repose, and coefficient of friction. These properties were used in selecting appropriate design parameters for the cleaning unit.

The performance of the machine was evaluated using cleaning efficiency, output capacity, grain loss, and blower efficiency. The soybean used for testing had an initial moisture content of 12% (wet basis).

The developed machine achieved an average cleaning efficiency of 93%, output capacity of 37.20 kg/h, grain loss of 2.67%, and blower efficiency of 60%.

5.2 Conclusion

Based on the results obtained from the development and testing of the pedal-operated soybean cleaning machine, the following conclusions were drawn:

1. A functional pedal-operated soybean cleaning machine was successfully developed using locally available materials.

2. The machine effectively removed impurities from soybean through the combined action of mechanical screening and air aspiration.

3. The physical properties of soybean determined in this study provided essential design information for the cleaning machine components.

4. The machine achieved satisfactory cleaning performance with an average cleaning efficiency of 93%.

5. The machine provides a low-cost alternative to electrically powered soybean cleaners and is suitable for small-scale agricultural applications.

6. The pedal-operated system eliminates dependence on electricity, making the machine appropriate for rural farming communities.

5.3 Recommendations

Based on the findings of this study, the following recommendations are made:

1. Further improvement should be made to increase the output capacity of the

machine while maintaining high cleaning efficiency.

2. The blower design should be optimized to improve airflow generation and reduce energy losses.

3. Different sieve sizes should be tested to determine the optimum size for different soybean varieties.

4. Future studies should investigate the effect of operating speed, feed rate, and moisture content on machine performance.

5. The use of lightweight and corrosion-resistant materials should be considered to reduce machine weight and improve portability.

6. Further research should be conducted on developing adjustable settings that allow the machine to clean other legume crops.

REFERENCES

Adekanye, T. A., Osakpamwan, A. B., & Osaivbie, I. E. (2016). *Evaluation of a soybean threshing machine for small-scale farmers. Agricultural Engineering International: CIGR Journal, 18(2), 35–44.*

Akatuhurira, W., Tumutegyereize, P., Oluk, I., Baidhe, E., Kigozi, J., Mayanja, I., & Kivumbi, H. B. (2021). *Development and performance evaluation of a pedal operated seed cleaner (POS-Cleaner). SN Applied Sciences, 3(6), 675.*

Akatuhurira, W., Tumutegyereize, P., Oluk, I., Baidhe, E., Kigozi, J., Mayanja, I., & Kivumbi, H. B. (2021). *Development and performance evaluation of a pedal-operated seed cleaner (POS-Cleaner).* SN Applied Sciences, 3(675), 1–14. https://doi.org/10.1007/s42452-021-04612-6

American Society of Agricultural and Biological Engineers. (2015). ASABE standards: *Agricultural machinery management data (ASAE D497.7). ASABE.*

Aradwad, P. P., Kumar, A., Sinha, J. P., & Yadav, R. S. (2021). *Development and performance evaluation of a screen cleaner for on-farm grain processing. The Indian Journal of Agricultural Sciences,* 91(5), 736–741. https://doi.org/10.56093/ijas.v91i5.112985

Bamgboye, A. I., & Mofolasayo, A. S. (2006). *Performance evaluation of a soybean threshing machine for small-scale farmers. Agricultural Engineering International: CIGR Journal, 8, 1–10.*

Food and Agriculture Organization. (2020). *Labour-saving technologies and practices: Manual and motorised cleaning of grains and pulses.*

Food and Agriculture Organization. (1992). *Technology of production of edible flours and protein products from soybeans.*

Hurburgh, C. R., Jr., Buresch, J., & Rippke, G. (1996). *Aspiration cleaning of soybeans. Applied Engineering in Agriculture, 12(5), 585–586.*

Henderson, S. M., & Perry, R. L. (1980). *Agricultural process engineering (3rd ed.). AVI Publishing Company.*

International Organization for Standardization. (2009). ISO 7971-3: *Cereals—Determination of bulk density, called mass per hectolitre—Part 3: Routine method. International Organization for Standardization.*

International Organization for Standardization. (2010). ISO 712: *Cereals and cereal products—Determination of moisture content—Reference method. International Organization for Standardization.*

Kabir, A. A., & Fedele, O. K. (2018). *A review of shelling, threshing, de-hulling, and decorticating machines. Open Access Journal of Agricultural Research, 3(1),*

1–10. https://doi.org/10.23880/OAJAR-16000148

Khurmi, R. S., & Gupta, J. K. (2019). *A textbook of machine design (25th ed.). S. Chand Publishing.*

Lin, W. (1996). *Costs and benefits of cleaning U.S. soybeans: Overview and implications. United States Department of Agriculture, Economic Research Service.*

Mabie, H. H., & Reinholtz, C. F. (1987). *Mechanisms and dynamics of machinery (4th ed.). John Wiley & Sons.*

Mayanja, I. K., Kigozi, J., Kawongolo, J. B., & Brumm, T. J. (2018). *Design, fabrication, and testing of pedal-operated maize grain cleaner. Journal of Advances in Food Science & Technology, 5(3), 105–111.*

Mohsenin, N. N. (1986). *Physical properties of plant and animal materials: Structure, physical characteristics and mechanical properties (2nd ed.). Gordon and Breach Science Publishers.*

Ojomo, A. O., & Fawole, O. O. (2014). *Design and performance evaluation of a grain cleaning machine. Journal of Agricultural Engineering and Technology, 22(1), 45–56.*

Satimehin, A. A., Omofunmi, O. E., & Uzua, M. M. (2019). *Status of postharvest processing operations of soybeans in rural Nigeria: A case study of Benue North-West Senatorial Zone. LAUTECH Journal of Engineering and Technology, 13(2), 34–43.*

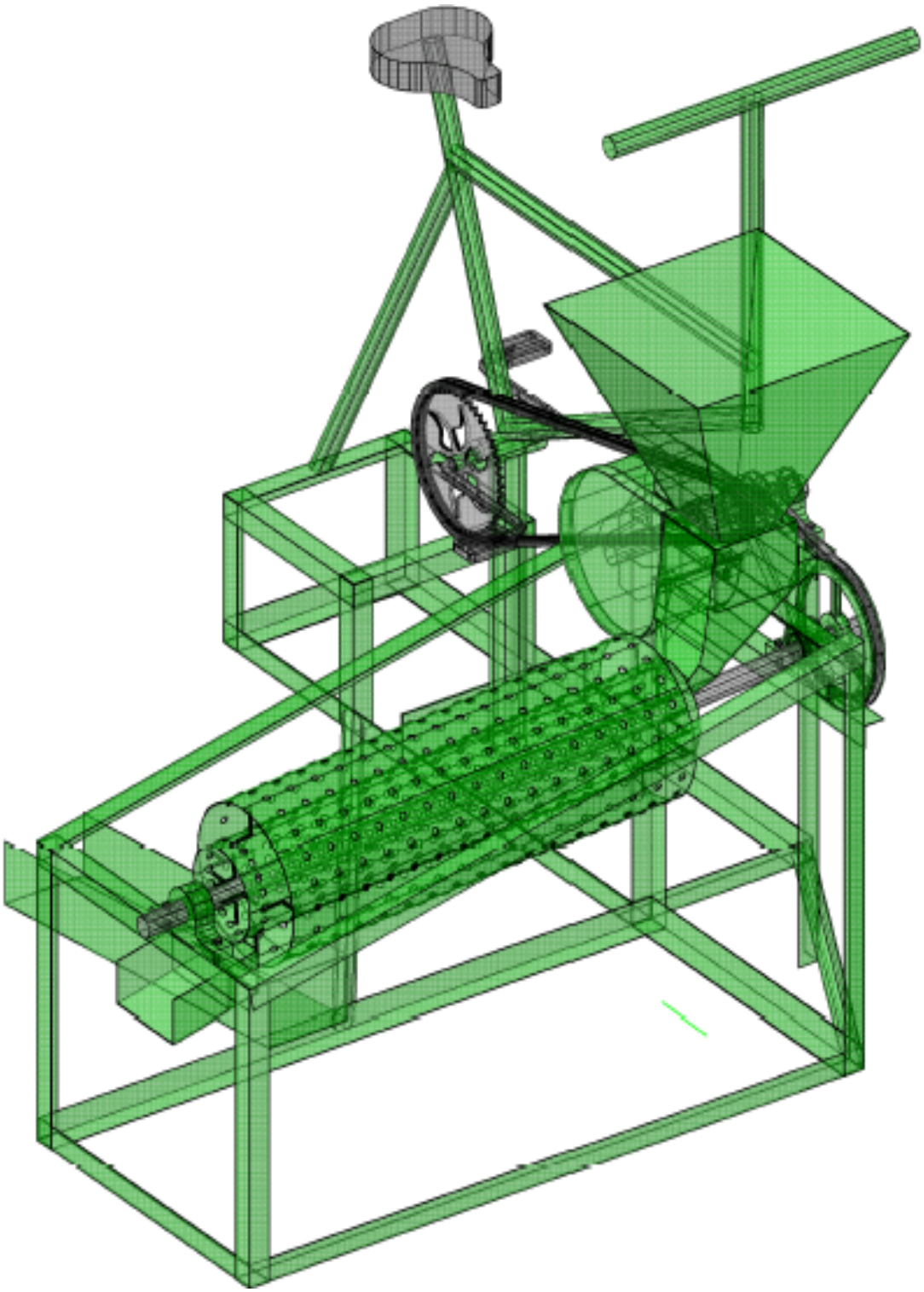
Sharma, P. C., & Aggarwal, D. K. (2018). *Machine design (12th ed.). S. K. Kataria & Sons.*

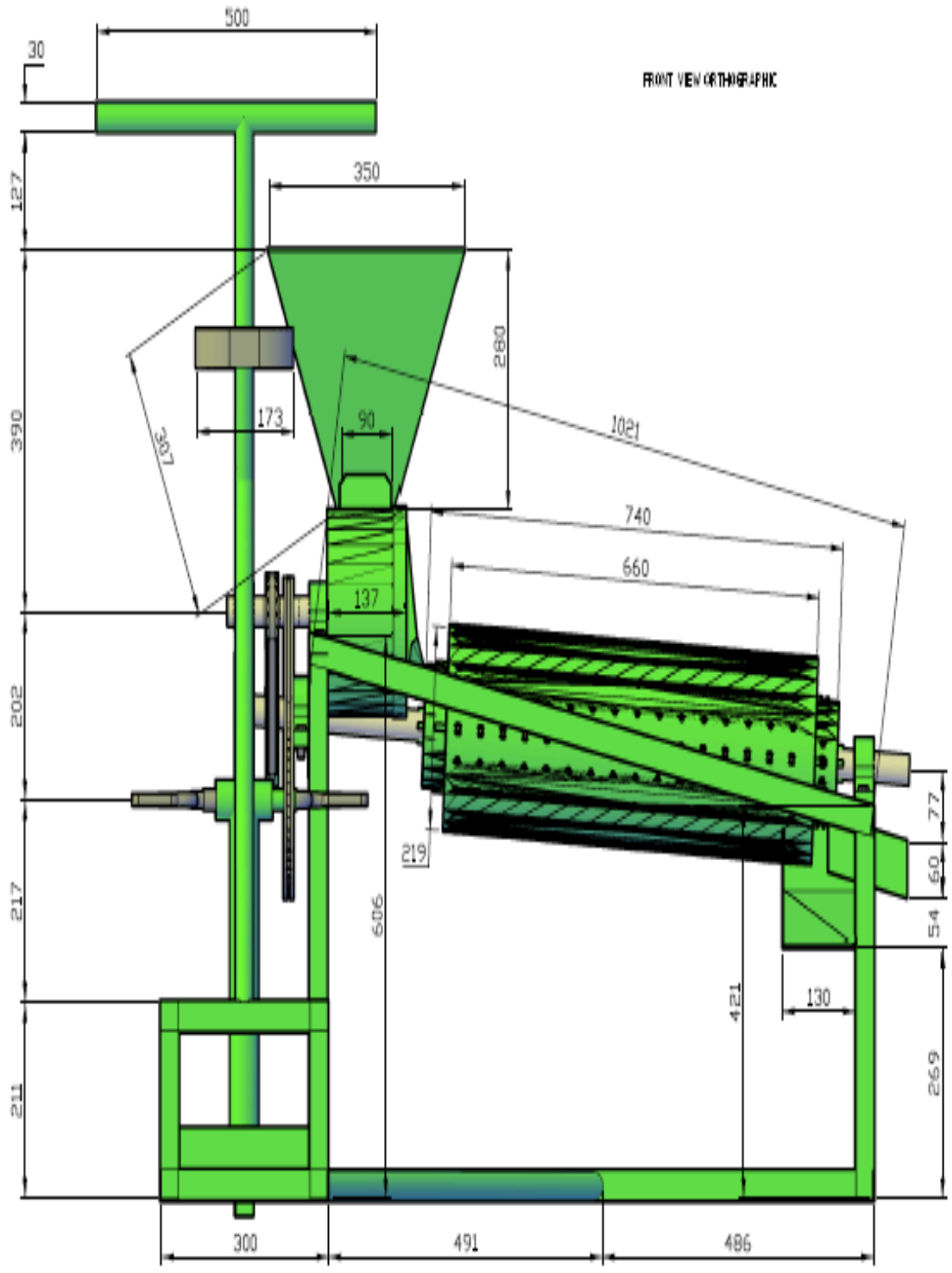
Singh, K. K., Goswami, T. K., & Raghavan, G. S. V. (1993). *Physical properties of grain legumes. Journal of Agricultural Engineering Research, 55(2), 89–98.* https://doi.org/10.1006/jaer.1993.1034

Toomer, O. T., Oviedo, E. O., Ali, M., Patino, D., Joseph, M., Frinsko, M., Vu, T., Maharjan, P., Fallen, B., & Mian, R. (2023). *Current agronomic practices, harvest, and post-harvest processing of soybeans (Glycine max): A review. Agronomy, 13(2), 427.* https://doi.org/10.3390/agronomy13020427

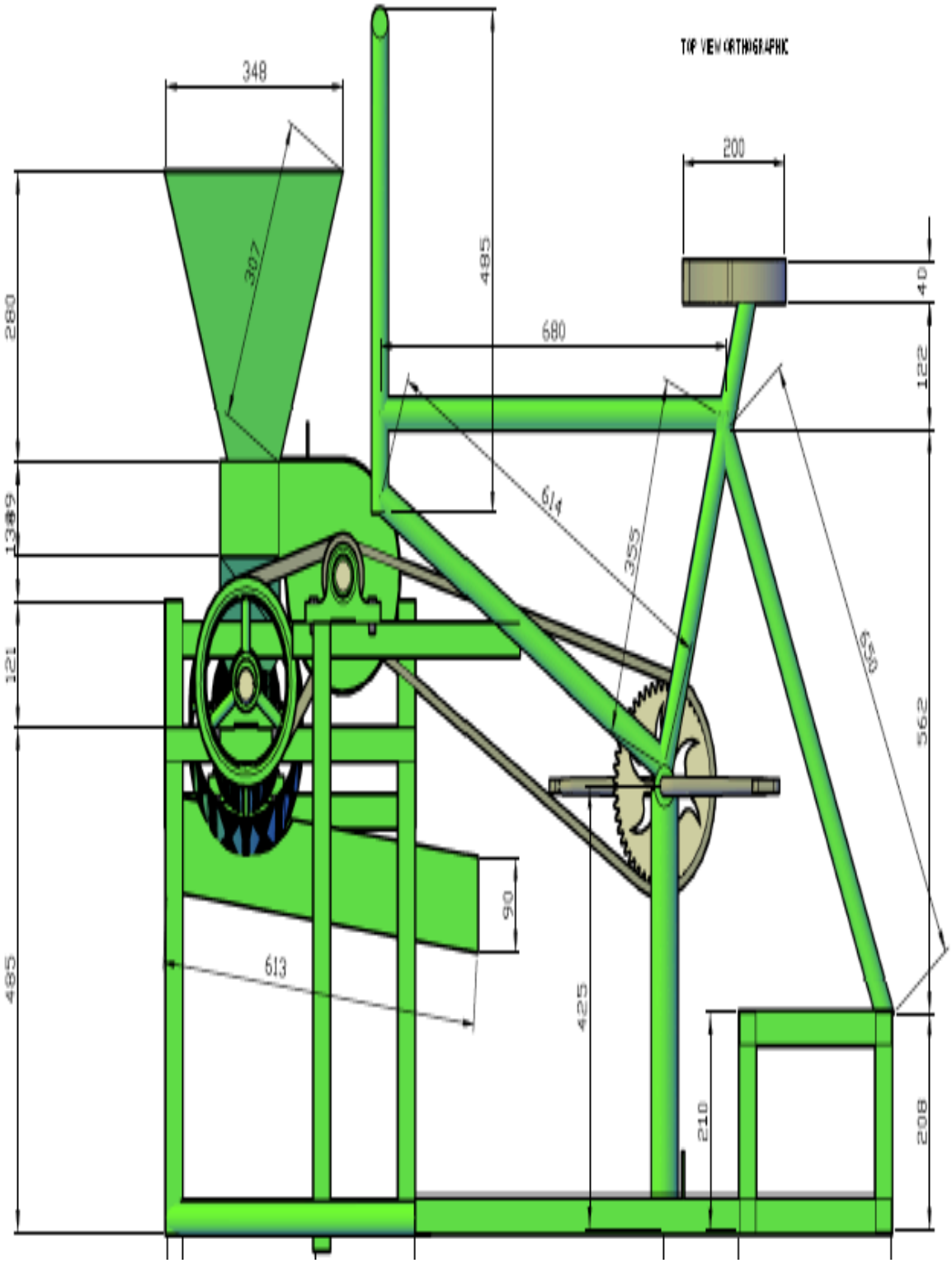
Zhang, X., Ji, S., Chen, L., Zhou, M., & Xia, X. (2026). *A review of mechanized harvesting, threshing, and cleaning devices for pulses. Agriculture, 16(10), 1051.* https://doi.org/10.3390/agriculture16101051

APPENDICES




FRONT VIEW ORTHOGRAPHIC
500
30
350
127
280
390
173
90
367
1021
740
660
137
202
77
219
60
54
217
606
421
130
269
211
300
491
486

Front View Orthographic (all dimension are in mm)



Top View Orthographic (all dimensions are in mm)



Group Picture